

Ryan Dwayne Koym

Senior Software Engineer

✉ ryankoy708@gmail.com ☎ +1 214 556 6262 📍 Fort Worth, TX, 76107

🌐 <http://www.linkedin.com/in/ryan-koym-87b914349>

PROFILE

Experienced Senior Java Full Stack Engineer with over 12 years of hands-on expertise in Java, .NET, Microsoft Azure, AWS, and modern frontend technologies. Specialize in creating scalable, high-performance, and secure cloud-native applications, with a deep focus on microservices architecture. With a strong background in building solutions for industries like healthcare and finance, I bring proficiency in CI/CD pipelines, event streaming, and cloud migration strategies. Well-versed in developing both backend and frontend systems, ensuring compliance with standards such as HIPAA, PCI DSS, and GDPR. Experienced team leader and mentor with a track record of achieving success through Agile practices.

PROFESSIONAL EXPERIENCE

Mayo Clinic

Senior Java Engineer

Sep 2020 – Present | Rochester, MN, Remote

Overview:

Led the modernization of Mayo Clinic's Florida Patient Scheduling Project, transitioning it to a scalable, microservices-based architecture. Delivered secure, high-performance solutions leveraging Java, Spring Boot, AWS services such as EC2, Lambda, S3, and MSK, ensuring regulatory compliance and enhancing system efficiency.

Key Contributions

System Architecture & Leadership

- Spearheaded the architecture and development of the Florida Patient Scheduling Project using Java, Spring Boot (versions 2.x/3.x), and AWS services like Lambda and MSK, ensuring scalability and fault tolerance.
- Introduced Kotlin to enhance code readability and reduce boilerplate, improving developer productivity by 20%.
- Led the migration from on-premises infrastructure to AWS cloud, leveraging Amazon MSK for real-time event streaming.
- Collaborated with cross-functional teams, mentoring junior developers, and implementing Agile practices to enhance team efficiency and delivery.
- Defined coding standards, conducted code reviews, and ensured adherence to healthcare compliance standards such as HIPAA.

Backend Development

- Designed and implemented over 30 microservices using Spring Boot and AWS Lambda for reliable and scalable communication between distributed services.
- Developed secure RESTful APIs and GraphQL to streamline patient scheduling data access, achieving a 25% reduction in latency and 30% faster response times.
- Integrated Spring Data JPA for database operations with AWS RDS (PostgreSQL/MySQL), ensuring optimized data retrieval and storage.

Frontend Development with React

- Built and maintained secure, HIPAA-compliant React applications, delivering intuitive interfaces for healthcare providers and patients.
- Developed reusable UI components in React, streamlining development and ensuring consistency across the application.
- Enhanced user experience by integrating real-time data visualization dashboards using WebSockets for updates on patient scheduling status.

AWS Cloud Integration

- Implemented AWS Lambda and Amazon S3 for serverless workflows, optimizing resource utilization and automating critical processes.
- Leveraged AWS RDS for scalable, secure database storage, and AWS IAM for secure role-based authentication.
- Implemented AWS CloudWatch for monitoring and logging, enabling rapid issue identification and resolution.

Data Integration and Performance Optimization

- Migrated legacy scheduling data from on-premises systems to AWS RDS using AWS DMS (Database Migration Service), achieving zero data loss and maintaining 99.9% uptime.
- Enhanced performance with AWS ElastiCache for Redis, reducing data retrieval times by 35% for frequently accessed scheduling records.
- Designed a robust ETL pipeline with AWS Glue and Apache Kafka, streamlining the processing of large-scale healthcare datasets.

Testing, CI/CD, and Monitoring

- Led integration testing using JUnit and Mockito, ensuring interoperability across all microservices.
- Established CI/CD pipelines with AWS CodePipeline, automating build, test, and deployment processes to production environments.
- Integrated AWS CloudWatch and ELK Stack (Elasticsearch, Logstash, Kibana) for comprehensive logging and monitoring.

Security and Compliance

- Enforced end-to-end encryption with AWS KMS and TLS/SSL protocols to secure sensitive patient data.
- Designed and implemented robust auditing mechanisms to ensure compliance with healthcare regulations and standards.

Achievements

- Delivered a modernized patient scheduling system that improved operational efficiency by 40% and reduced system downtime to less than 2 hours annually.
- Successfully migrated 10+ million scheduling records to AWS RDS, enhancing data availability and enabling faster clinical decision-making.
- Introduced Kotlin into the tech stack, reducing code complexity and boosting development speed by 15%-20%.
- Led a team of 15+ engineers, scaling the team to 25+ members while improving project delivery speed by 10%.
- Facilitated the transition to AWS MSK for real-time streaming, resulting in enhanced scalability and seamless integration with the broader big data ecosystem for advanced analytics and reporting.

Home Depot
Senior Software Engineer

Jan 2016 – Aug 2020 | Irving, TX

Overview:

Contributed to the development and optimization of Home Depot's e-commerce platform, a high-demand retail system supporting millions of daily users. Focused on building scalable

microservices, responsive interfaces, and cloud-based solutions to ensure seamless customer experiences during peak shopping periods.

Key Contributions

Overview:

Led the development and optimization of **Home Depot's e-commerce platform**, ensuring **scalability, high availability, and real-time data processing**. Designed and built **cloud-native microservices** using **Java, Spring Boot, AWS, and GCP**, optimizing system performance during peak shopping seasons.

Key Contributions:

Cloud-Native Architecture (AWS & GCP)

- Designed and implemented **microservices** using **Java, Spring Boot, and Spring Cloud**, deployed across **AWS (EKS, Lambda)** and **GCP (Cloud Run, GKE)** for **scalability and fault tolerance**.
- **Migrated legacy services** to a hybrid **AWS-GCP** environment, leveraging **Google Cloud Storage (GCS)** for durable, cost-effective data retention.
- Automated **infrastructure deployment** using **Terraform** for multi-cloud consistency.

Event-Driven Architecture & Kafka

- Implemented a **real-time order processing system** using **Kafka (Amazon MSK + Google Pub/Sub)**, ensuring seamless inventory updates and transaction reliability.
- Developed **event-driven microservices** that processed **high-volume transactions**, ensuring a **99.99% uptime SLA**.
- Integrated **Kafka Streams** for **real-time analytics** on order trends and fraud detection.

E-Commerce Optimization & High Availability

- Developed **RESTful APIs** for **inventory, pricing, and order management**, reducing **API response times by 20%** through optimized database queries and caching strategies.
- Leveraged **AWS ElastiCache (Redis)** and **Google Memorystore** to improve application performance.
- Designed **fault-tolerant systems** by implementing **circuit breakers (Resilience4j)** and multi-region deployments across **AWS and GCP**.

CI/CD & DevOps

- Automated **build, testing, and deployment** pipelines with **Jenkins, GitHub Actions, and AWS CodePipeline**, ensuring smooth, zero-downtime releases.
- Implemented **containerization (Docker, Kubernetes)** for service scalability and seamless deployment across **AWS EKS & GCP GKE**.
- Monitored applications with **AWS CloudWatch, GCP Operations Suite, and ELK Stack (Elasticsearch, Logstash, Kibana)** for centralized logging and issue detection.

Cross-Team Collaboration & Leadership

- Partnered with **PMs, UX designers, and data engineers** to develop **real-time analytics dashboards** for tracking sales trends using **BigQuery and AWS QuickSight**.
- Mentored junior developers on **Spring Boot, Kafka, and cloud best practices**, fostering a culture of **continuous learning and innovation**.

Achievements:

- Reduced **page load times by 30%** during high-traffic events like **Black Friday**, improving customer experience.
- Optimized **Kafka event streaming**, reducing **order processing time by 20%**.
- Successfully **migrated 50M+ records** from on-prem to **AWS RDS & Google BigQuery**, ensuring **real-time analytics** and improved reporting accuracy.
- Increased **e-commerce platform scalability by 50%**, ensuring seamless transactions for **millions of users daily**.

About Digital IQ:

Digital IQ, Inc. is a software development company specializing in delivering secure, scalable, and domain-specific solutions across finance and healthcare. Leveraging modern technologies, the company focuses on building compliant platforms for regulated industries.

Overview of Projects

Healthcare Domain: HIPAA-Compliant Patient Data Platform

Developed and enhanced a responsive patient data management system to streamline clinical workflows and ensure compliance with healthcare regulations like HIPAA. Focused on real-time data processing, multi-platform support, and secure interfaces for patient and administrator access.

Finance Domain: Insurance Policy Management System

Contributed to a robust insurance policy management platform, enabling efficient policy creation, claims processing, and reporting. Implemented features to enhance operational efficiency while ensuring data security and regulatory compliance.

Key Contributions

Backend Development

- Built scalable microservices using **Java** and **Spring MVC** (4.x), enabling efficient processing of healthcare records and insurance policies.
- Designed **SOAP** and **RESTful APIs** with **Spring Web Services** and **Spring Boot**, ensuring seamless integration between third-party systems and internal applications.
- Developed robust data processing pipelines for healthcare and insurance workflows, leveraging **Hibernate** and **MySQL/SQL Server** for secure, optimized database interactions.
- Implemented business logic for insurance claim validation and healthcare data transformation using **Java Streams** and **stored procedures**.

Frontend Development

- Developed user-friendly, responsive interfaces using **AngularJS** (1.x), **HTML5**, **CSS3**, and **Bootstrap**, ensuring compatibility across devices for healthcare professionals and insurance agents.
- Enhanced user experience by integrating real-time data updates and dashboards using **WebSockets**, improving data visibility and decision-making.

Data Management & Reporting

- Utilized **MySQL/PostgreSQL** to optimize queries, ensuring fast data retrieval for high-traffic applications.
- Built comprehensive reports with **JasperReports** or **iReport** providing actionable insights for administrators in healthcare and finance.

Testing & Version Control

- Conducted unit testing with **JUnit** and **Mockito**, ensuring the reliability of core application components.
- Employed **Git** and **Jenkins** for version control and continuous integration, maintaining code integrity and enabling automated testing and deployment pipelines.

Performance Optimization

- Achieved significant performance gains by optimizing database queries and implementing caching mechanisms using **Redis** and **Ehcache**.
- Reduced API response times by 20% through code refactoring and performance tuning across healthcare and finance platforms.

Security & Compliance

- Enforced HIPAA compliance for healthcare applications by implementing encryption with

TLS/SSL and securing sensitive data in transit and at rest.

- Designed role-based access control (RBAC) systems for insurance applications, ensuring secure access for different user roles.

Cloud Integration (Emerging Exposure)

- Developed foundational experience with **AWS services** (such as **S3**, **EC2**, **RDS**) for secure file storage, deployment, and data management.
- Implemented **AWS IAM** for streamlined user authentication and access control management.

EDUCATION

Bachelor's Degree, Computer Science
University of North Texas

2009 – 2012

SKILLS

Programming Languages

Java, C#, JavaScript, TypeScript, C#, Python, SQL, HTML, CSS, XML, XSLT, PowerShell

Frameworks and Libraries

Spring Framework (Boot, Cloud, Security), Spring MVC, .NET Core, ASP.NET MVC, Hibernate, React, Angular, jQuery, AWS SDK, JPA

Microservices

Microservices architecture, RESTful API, gRPC, AWS Lambda, AWS SNS, AWS SQS, Service Bus, Kafka (Amazon MSK), Event-driven architecture, Caching Mechanism, Performance Optimization, PCI DSS, HIPAA Compliance

Databases

MySQL, PostgreSQL, SQL Server, Cassandra, MongoDB, Redis, AWS DynamoDB, AWS RDS

Cloud Services

Amazon EC2, AWS Lambda, Amazon S3, Amazon RDS, Amazon MSK, ElasticSearch, AWS IAM, Amazon SQS, AWS SNS, AWS CloudWatch, AWS ECS, AWS EKS

DevOps and CI/CD

GitHub Actions, Jenkins, Docker, Kubernetes, GitLab CI/CD, Terraform, Ansible, AWS CodePipeline, AWS CodeDeploy

Testing, Monitoring, and Logging

JUnit, TestNG, Mockito, Selenium, Postman, AWS CloudWatch, AWS X-Ray, ELK Stack (Elasticsearch, Logstash, Kibana), Prometheus, Grafana

Web Development

RESTful APIs, GraphQL, React, Angular, Bootstrap, Tailwind CSS, Material-UI, Responsive Design

Tools and Technologies

Visual Studio Code, IntelliJ IDEA, Git, JIRA, Swagger, Postman, AWS Management Console

Methodologies

Agile, Scrum, Kanban, Test-Driven Development (TDD), Behavior-Driven Development (BDD)

